

Aim:

- Viewing and inspecting DataFrames
- Filtering and subsetting data using conditions
- Descriptive statistics: measures of central tendency (mean, median, mode) and measures of dispersion (range, variance, standard deviation)

CODE:

```
import pandas as pd
df = pd.read_csv('/content/test_Y3wMUE5_7gLdaTN.csv')
print(df.head())
```

```

Loan_ID  Gender  Married  Dependents  Education  Self_Employed  \
0  LP001015   Male     Yes           0    Graduate             No
1  LP001022   Male     Yes           1    Graduate             No
2  LP001031   Male     Yes           2    Graduate             No
3  LP001035   Male     Yes           2    Graduate             No
4  LP001051   Male     No            0  Not Graduate             No

ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term  \
0              5720                  0         110.0             360.0
1              3076             1500         126.0             360.0
2              5000             1800         208.0             360.0
3              2340             2546         100.0             360.0
4              3276                  0          78.0             360.0

Credit_History  Property_Area
0              1.0          Urban
1              1.0          Urban
2              1.0          Urban
3              NaN          Urban
4              1.0          Urban
```

```
print(df.tail())
```

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```
Loan_ID Gender Married Dependents Education Self_Employed \
362 LP002971 Male Yes 3+ Not Graduate Yes
363 LP002975 Male Yes 0 Graduate No
364 LP002980 Male No 0 Graduate No
365 LP002986 Male Yes 0 Graduate No
366 LP002989 Male No 0 Graduate Yes

ApplicantIncome CoapplicantIncome LoanAmount Loan_Amount_Term \
362 4809 1777 113.0 360.0
363 4158 709 115.0 360.0
364 3250 1993 126.0 360.0
365 5000 2393 158.0 360.0
366 9200 0 98.0 180.0

Credit_History Property_Area
362 1.0 Urban
363 1.0 Urban
364 NaN Semiurban
365 1.0 Rural
366 1.0 Rural
```

print(df.info())

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID                367 non-null   object
1   Gender                 356 non-null   object
2   Married                367 non-null   object
3   Dependents             357 non-null   object
4   Education              367 non-null   object
5   Self_Employed          344 non-null   object
6   ApplicantIncome        367 non-null   int64
7   CoapplicantIncome      367 non-null   int64
8   LoanAmount             362 non-null   float64
9   Loan_Amount_Term       361 non-null   float64
10  Credit_History         338 non-null   float64
11  Property_Area          367 non-null   object
dtypes: float64(3), int64(2), object(7)
memory usage: 34.5+ KB
None
```

print(df.describe())

```
ApplicantIncome CoapplicantIncome LoanAmount Loan_Amount_Term \
count  367.000000  367.000000  362.000000  361.000000
mean    4805.599455  1569.577657  136.132597  342.537396
std     4910.685399  2334.232099   61.366652   65.156643
min       0.000000    0.000000   28.000000    6.000000
25%     2864.000000    0.000000  100.250000  360.000000
50%     3786.000000  1025.000000  125.000000  360.000000
75%     5060.000000  2430.500000  158.000000  360.000000
max     72529.000000 24000.000000  550.000000  480.000000

Credit_History
count  338.000000
mean    0.825444
std     0.380150
min     0.000000
25%     1.000000
50%     1.000000
75%     1.000000
max     1.000000
```

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```
print(df.columns)
```

```
Index(['Loan_ID', 'Gender', 'Married', 'Dependents', 'Education',  
       'Self_Employed', 'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',  
       'Loan_Amount_Term', 'Credit_History', 'Property_Area'],  
      dtype='object')
```

```
categorical_cols = ['Gender', 'Married', 'Dependents', 'Education',  
                   'Self_Employed', 'Property_Area']
```

```
for col in categorical_cols:
```

```
    unique_vals = df[col].unique()
```

```
    print(f"\nUnique values in '{col}': {unique_vals}")
```

```
↳
```

```
Unique values in 'Gender': ['Male' 'Female' nan]
```

```
Unique values in 'Married': ['Yes' 'No']
```

```
Unique values in 'Dependents': ['0' '1' '2' '3+' nan]
```

```
Unique values in 'Education': ['Graduate' 'Not Graduate']
```

```
Unique values in 'Self_Employed': ['No' 'Yes' nan]
```

```
Unique values in 'Property_Area': ['Urban' 'Semiurban' 'Rural']
```

```
df['Dependents'] = df['Dependents'].replace('3+', 3)
```

```
df['Dependents'] = pd.to_numeric(df['Dependents'],
```

```
errors='coerce').astype('Int64')
```

```
print("\nValue counts in 'Dependents' after conversion:")
```

```
print(df['Dependents'].value_counts(dropna=False))
```

```
Value counts in 'Dependents' after conversion:
```

```
Dependents
```

```
0      200
```

```
2       59
```

```
1       58
```

```
3       40
```

```
<NA>    10
```

```
Name: count, dtype: Int64
```

```
cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']
```

```
for col in cols:
```

```
    print(f"\nStatistics for '{col}':")
```

```
    print(f"Mean: {df[col].mean():.2f}")
```

```
    print(f"Median: {df[col].median():.2f}")
```

```
    print(f"Mode: {df[col].mode().values}")
```

```
    print(f"Range: {df[col].max() - df[col].min():.2f}")
```

```
    print(f"Variance: {df[col].var():.2f}")
```

```
    print(f"Standard Deviation: {df[col].std():.2f}")
```

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```
Statistics for 'ApplicantIncome':  
Mean: 4805.60  
Median: 3786.00  
Mode: [3500 5000]  
Range: 72529.00  
Variance: 24114831.09  
Standard Deviation: 4910.69  
  
Statistics for 'CoapplicantIncome':  
Mean: 1569.58  
Median: 1025.00  
Mode: [0]  
Range: 24000.00  
Variance: 5448639.49  
Standard Deviation: 2334.23  
  
Statistics for 'LoanAmount':  
Mean: 136.13  
Median: 125.00  
Mode: [150.]  
Range: 522.00  
Variance: 3765.87  
Standard Deviation: 61.37
```

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Result:

Thus the EDA-Data Inspection and Analysis is done successfully.